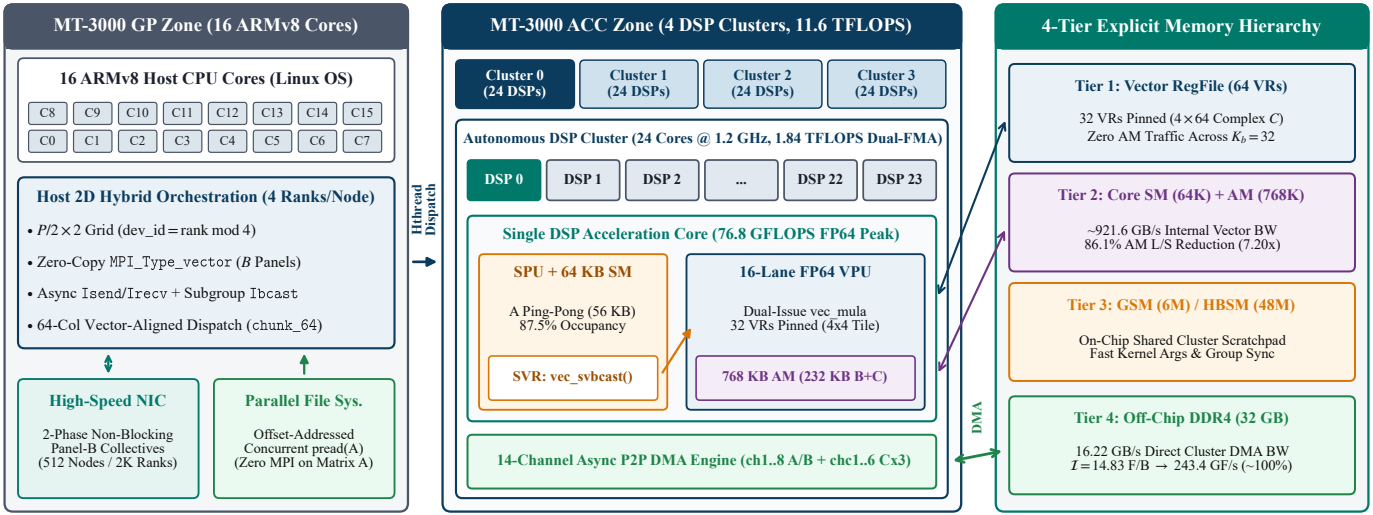

MT-3000 GP Zone (16 ARMv8 Cores)
16 ARMv8 Host CPU Cores (Linux OS)
C8 C9 C10 C11 C12 C13 C14 C15
C0 C1 C2 C3 C4 C5 C6 C7
Host 2D Hybrid Orchestration (4 Ranks/Node)
• P/2 × 2 Grid (dev_id = rank mod 4)
• Zero-Copy MPI_Type_vector (B Panels)
• Async Isend/Irecv + Subgroup Ibcast
• 64-Col Vector-Aligned Dispatch (chunk_64)
High-Speed NIC
2-Phase Non-Blocking
Panel-B Collectives
(512 Nodes / 2K Ranks)
Parallel File Sys.
Offset-Addressed
Concurrent pread(A)
(Zero MPI on Matrix A)
Hthread Dispatch
MT-3000 ACC Zone (4 DSP Clusters, 11.6 TFLOPS)
Cluster 0 (24 DSPs)
Cluster 1 (24 DSPs)
Cluster 2 (24 DSPs)
Cluster 3 (24 DSPs)
Autonomous DSP Cluster (24 Cores @ 1.2 GHz, 1.84 TFLOPS Dual-FMA)
DSP 0
DSP 1
DSP 2
...
DSP 22
DSP 23
Single DSP Acceleration Core (76.8 GFLOPS FP64 Peak)
SPU + 64 KB SM
A Ping-Pong (56 KB)
87.5% Occupancy
SVR: vec_svbcast()
16-Lane FP64 VPU
Dual-Issue vec_mula
32 VRs Pinned (4x4 Tile)
768 KB AM (232 KB B+C)
14-Channel Async P2P DMA Engine (ch1..8 A/B + ch1..6 Cx3)
DMA
4-Tier Explicit Memory Hierarchy
Tier 1: Vector RegFile (64 VRs)
32 VRs Pinned (4 × 64 Complex C)
Zero AM Traffic Across $K_b = 32$
Tier 2: Core SM (64K) + AM (768K)
~921.6 GB/s Internal Vector BW
86.1% AM L/S Reduction (7.20x)
Tier 3: GSM (6M) / HBSM (48M)
On-Chip Shared Cluster Scratchpad
Fast Kernel Args & Group Sync
Tier 4: Off-Chip DDR4 (32 GB)
16.22 GB/s Direct Cluster DMA BW
I = 14.83 F/B → 243.4 GF/s (~100%)